

ML Assignment 3

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Section A (Theoretical)

Ques 1:

2.1) $x = [1, 2, 3]$
 $y = [3, 4, 5]$

Let $w_1 = 0.5$
 $b_1 = 0.1$
 $w_2 = 0.5$
 $b_2 = 0.2$

$\lambda = 0.01$

ReLU $\Rightarrow h_i = \max(0, h_i)$

For 1st hidden layer

$$h_1 = \max(0, 0.5 + 0.1) = 0.6$$
$$h_2 = \max(0, 0.5 \times 2 + 0.1) = 1.1$$
$$h_3 = \max(0, 0.5 \times 3 + 0.1) = 1.6$$
$$h = [0.6, 1.1, 1.6]$$

For output layer

$$\text{output}_1 = 0.5(0.6) + 0.2 = 0.5$$
$$\text{output}_2 = 0.5(1.1) + 0.2 = 0.75$$
$$\text{output}_3 = 0.5(1.6) + 0.2 = 1$$
$$\text{output} = [0.5, 0.75, 1]$$

$$\begin{aligned}
 MSE &= \frac{1}{n} \sum_{i=1}^n (\text{output } i - y_i)^2 \\
 &= \frac{1}{3} ((2.5)^2 + (3.25)^2 + (4)^2) \\
 &= \frac{1}{3} (6.25 + 10.5625 + 16) \\
 &= \frac{1}{3} (32.8125)
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial \text{Loss}}{\partial w_1} &= \frac{2}{3} \sum_{i=1}^3 (\text{output } i - y_i) \cdot h_i \\
 &= \frac{2}{3} \times (-1.5 - 3.575 - 6.4) \\
 &= \frac{2}{3} \times -11.475 \\
 &\approx -7.65
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial \text{Loss}}{\partial b_2} &= \frac{2}{3} \sum_{i=1}^3 (\text{output } i - y_i) \\
 &= \frac{2}{3} (-2.5 - 3.25 - 4) \\
 &= \frac{2}{3} \times -9.75 \\
 &\approx -6.5
 \end{aligned}$$

$$\frac{\partial \text{Loss}}{\partial w_1} = \frac{2}{3} \sum_{i=1}^3 (\text{output}_i - y_i) \cdot w_2 \cdot n_i \cdot \text{ReLU}'(h_i)$$

$$= \frac{2}{3} \times (-1.25 - 3.25 - 6.0) \cdot (2.5) \cdot 1$$

$$= \frac{2}{3} \times -10.5 + 25.25 + 15.0 = 10$$

$$\approx -7$$

$$\frac{\partial \text{Loss}}{\partial b_1} = \frac{2}{3} \times \left(\sum_{i=1}^3 (\text{output}_i - y_i) \cdot w_i \cdot \text{ReLU}'(h_i) \right)$$

$$= \frac{2}{3} \times (-1.25 - 1.625 - 2)$$

$$= \frac{2}{3} \times (-4.875)$$

$$\approx -3.25$$

$$w_1 = w_1 - \alpha \frac{\partial \text{Loss}}{\partial w_1}$$

$$= 0.5 - 0.01(-0.7)$$

$$\approx 0.57$$

$$b_1 = b_1 - \alpha \frac{\partial \text{Loss}}{\partial b_1}$$

$$= 0.1 - 0.01(-3.25)$$

$$\approx 0.1325$$

$$w_2 = w_2 - \alpha \frac{\partial \text{Loss}}{\partial w_2}$$

$$= 0.5 - 0.01(-7.65)$$

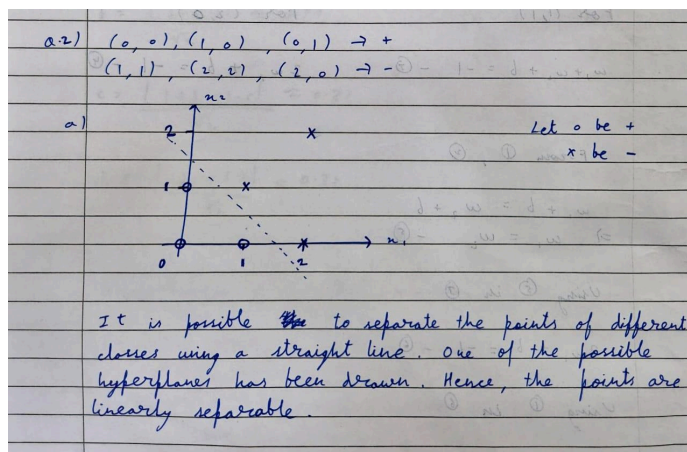
$$\approx 0.5765$$

$$b_2 = b_2 - \alpha \frac{\partial \text{Loss}}{\partial b_2}$$

$$= 0.2 - 0.01(-6.5)$$

$$\approx 0.265$$

Ques 2:



b) The potential support vectors are:

$(1,0), (0,1) \rightarrow +$
 $(1,1), (2,0) \rightarrow -$

$w_1 x_1 + w_2 x_2 + b \geq 1$ for class +
 $w_1 x_1 + w_2 x_2 + b \leq -1$ for class -

For $(1,0)$ For $(0,1)$

$w_1 + b = 1$ - ① $w_2 + b = 1$ - ②

For $(1,1)$ For $(2,0)$

$w_1 + w_2 + b = -1$ - ③ $2w_1 + b = -1$ - ④

From ①, ②

$w_1 + b = w_2 + b$
 $\Rightarrow w_1 = w_2$ - ⑤

Using ⑤ in ③

$2w_1 + b = -1$ - ⑥

Using ① in ⑥

$2w_1 + (1 - w_1) = -1$
 $\Rightarrow w_1 = -2$
 $w_2 = -2$
 $b = 3$

Maximum Margin Hyperplane:

$$-2x_1 - 2x_2 + 3 = 0$$

$$\Rightarrow x_1 + x_2 = \frac{3}{2}$$

checking the potential support vectors:

$$a = \frac{|1 - 1.5|}{\sqrt{2}} \approx 0.35$$

$$b = \frac{|1 - 1.5|}{\sqrt{2}} \approx 0.35$$

$$c = \frac{|1 + 1 - 1.5|}{\sqrt{2}} \approx 0.35$$

$$d = \frac{|2 - 1.5|}{\sqrt{2}} \approx 0.35$$

$$\text{weight Vector} \Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Bias} \Rightarrow -\frac{3}{2}$$

$$\text{support Vectors} \Rightarrow (1, 0), (0, 1), (1, 1), (2, 0)$$

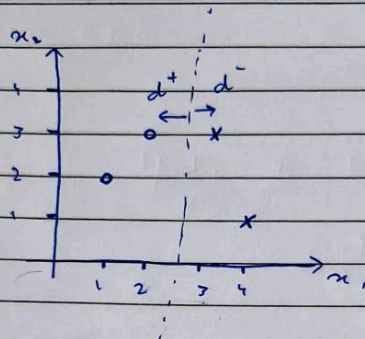
Ques 3:

Q.3)

$$\begin{aligned}(1, 2) &\rightarrow + \\ (2, 3) &\rightarrow + \\ (3, 3) &\rightarrow - \\ (4, 1) &\rightarrow -\end{aligned}$$

$$\begin{aligned}-2x_1 + 0x_2 &= -5 \quad (\text{Hyperplane}) \\ \Rightarrow x_1 &= \frac{5}{2}\end{aligned}$$

Let $\circ$ be +
x be -



a) Two types of margins : Theoretical, Empirical

will calculate both since not specified

① Theoretical $\Rightarrow \frac{1}{\|w\|}$

$$\begin{aligned}&= \frac{2}{\sqrt{(-2)^2 + 0^2}} = \frac{1}{2} \\ &= 0.5\end{aligned}$$

② Empirical $\Rightarrow d^+ + d^-$

$$\begin{aligned}&\Rightarrow \frac{|-2(2) + 5|}{2} + \frac{|-2(3) + 5|}{2} \\ &= 0.5 + 0.5 \\ &= 1\end{aligned}$$

$$b) \quad (1, 2) \Rightarrow d = \frac{|-2(1) + 5|}{2}$$

$$= 1.5$$

$$(2, 3) \Rightarrow d = \frac{|-2(2) + 5|}{2}$$

$$= 0.5$$

$$(3, 3) \Rightarrow d = \frac{|-2(3) + 5|}{2}$$

$$= 0.5$$

$$(4, 1) \Rightarrow d = \frac{|-2(4) + 5|}{2}$$

$$= 1.5$$

support vectors $\Rightarrow (2, 3), (3, 3)$

$$c) \quad b(x) = w_1 x_1 + w_2 x_2 + b$$

$$= (-2x_1 + 0 + 5)$$

$$\text{for } (1, 2), \quad b(x) = -2 + 5$$

$$= 3$$

$$b(x) > 0$$

$\Rightarrow$ The point is classified as $y = +1$

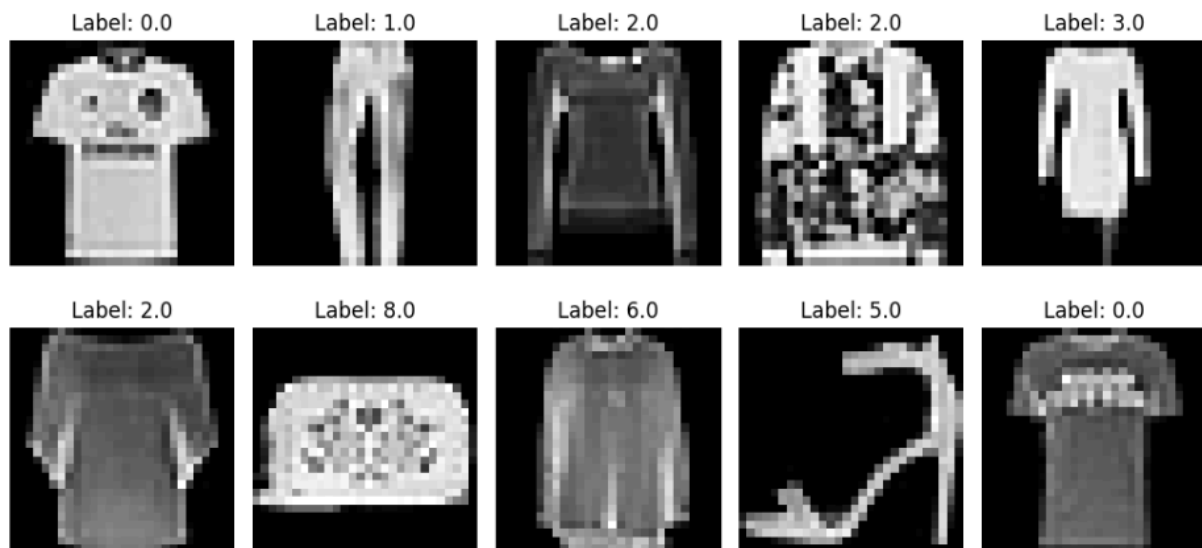
Section C (Algorithm Implementation using Packages)

Part 1

Performed the data split by taking the first 8000 images from the training dataset and the first 2000 images from the test dataset. Additionally, a validation set was created from the next 2000 images in the test dataset. There were no null values found and no categorical features were present.

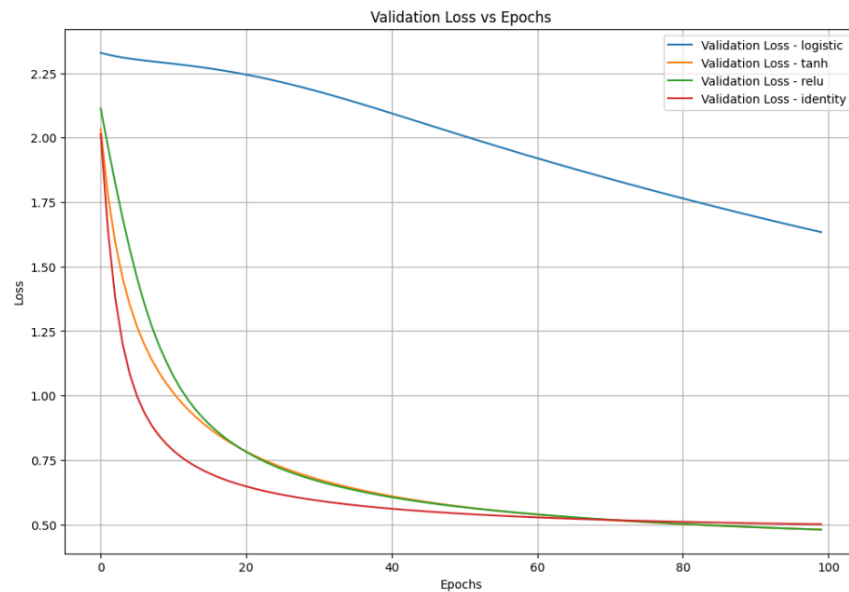
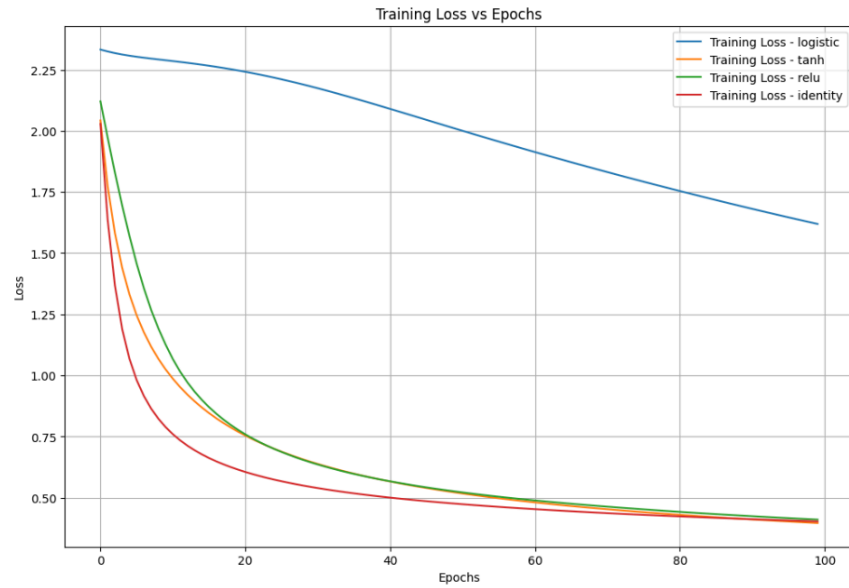
Each data point consists of 784 pixels corresponding to 28x28 grayscale images. The pixel values were normalized to the range of [0, 1] to enhance convergence during model training.

The first 10 images from the training set can be seen below as samples:



Part 2

MLP classifiers were trained according to the given configurations for logistic, tanh, relu, and identity activation functions. Below are the training loss vs. epochs and validation loss vs. epochs plots comparing the activation functions:



Below is the test loss for each activation function with relu being the best performing activation function achieving the best accuracy of 0.8365.

```
Test Loss for logistic activation function: 1.62324765694004
Test Accuracy for logistic activation function: 0.4405

Test Loss for tanh activation function: 0.472761303394661
Test Accuracy for tanh activation function: 0.8345

Test Loss for relu activation function: 0.48348738603887526
Test Accuracy for relu activation function: 0.8365

Test Loss for identity activation function: 0.49001333682487136
Test Accuracy for identity activation function: 0.828
```

Part 3

For the grid search, I used a 2-fold cross-validation to determine the optimal hyperparameters for the MLP classifier. The MLP was configured with three hidden layers of sizes (128, 64, 32) and employed the best activation function identified in Part 2 i.e. relu. accuracy was used as the scoring metric. Below are the grid search parameters:

- Solvers: ['adam', 'sgd', 'lbfgs']
- Learning Rates: [2e-3, 2e-5]
- Batch Sizes: [32, 64]

The parameters with the best accuracy of 0.8535 are:

```
Best Hyperparameters from Grid Search:
{'batch_size': 64, 'learning_rate_init': 0.002, 'solver': 'adam'}
Best Accuracy: 0.8534999999999999
```

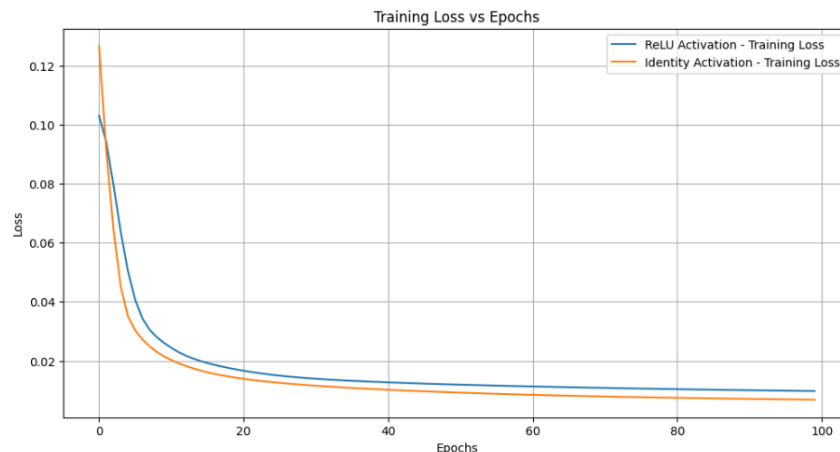
Part 4

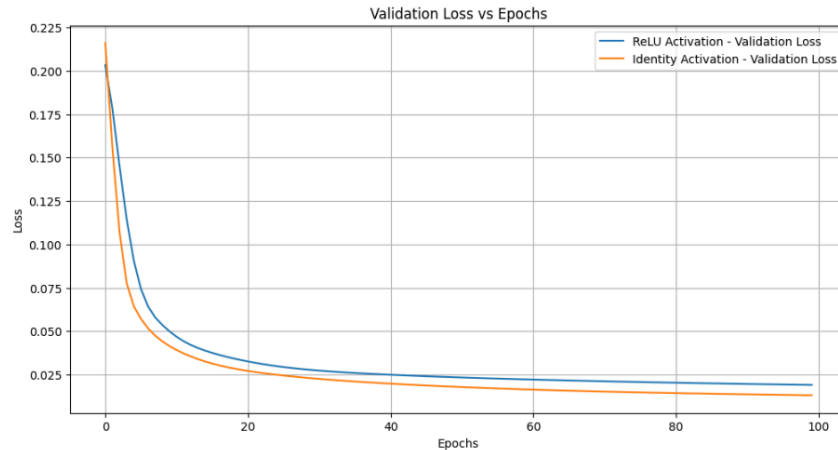
The layer sizes c, b, and a were chosen as follows:

- $c = \text{feature vector size} / 2 = 392$
- $b = c / 2 = 196$
- $a = b / 2 = 98$

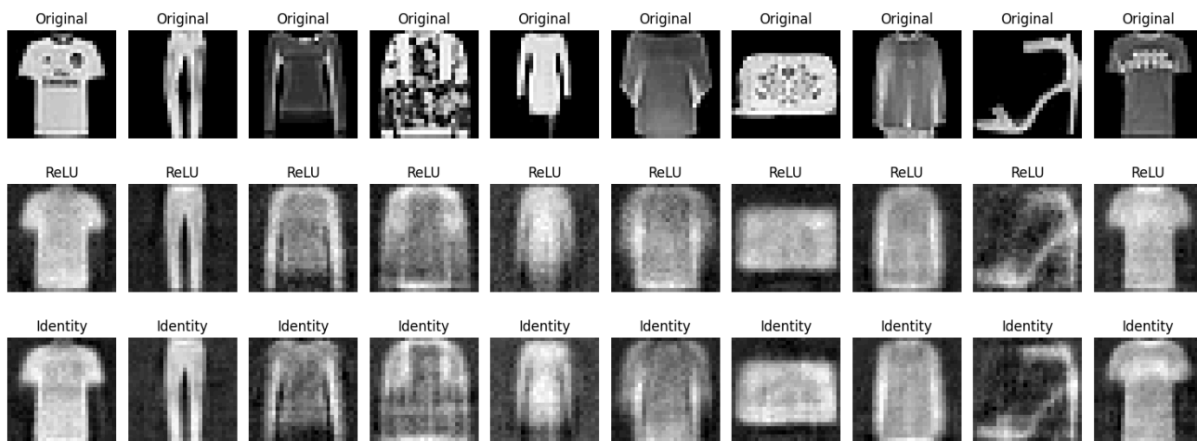
By halving the layer sizes in the earlier part of the network, the model learns key features at each stage, retaining only the most significant information. In the latter part of the network, the layer sizes are doubled, which allows the model to expand and refine the compressed features. This helps in reconstructing complex patterns leading to more accurate predictions.

Two models were trained according to the given configurations. The training loss vs. epochs and validation loss vs. epochs plots are shown below:





The Identity activation function provided better image reconstruction with better edge preservation and smoother images compared to ReLU as seen in the first 10 images from the training set below. It also achieved a lower loss than ReLU as seen in the plots above.



Part 5

A feature vector of size a (98 here) was extracted from the model trained in part 4. Since the model instances store the weights and biases for each layer, I use these to compute the output of the third hidden layer which is the feature vector of size a . This vector was then used as input for two new classifiers, each configured as per the requirements mentioned. The accuracy for these classifiers can be seen below.

```
Accuracy of ReLU Activation Classifier: 0.789
Accuracy of Identity Activation Classifier: 0.7825
```

In part 5, the classifier performs slightly lower at 0.789 accuracy compared to 0.8365 in part 2, but it's still a pretty decent classifier. This is because it uses a feature vector that captures key information from the images making it possible to classify well even with less input data. The

simpler model in part 5 is also less complex which helps it generalize better and avoid overfitting making it more efficient.